Matrices

Matrices

Stores 2-D values.

```
matrix(
  c(80, 90, 85, 85, 90, 90, 85, 100),
  nrow = 4,
  ncol = 2
)
```

```
column_1 <- c(80, 90, 85, 85)
column_2 <- c(90, 90, 85, 100)

cbind(column_1, column_2)
```

Matrices

Indexing is similar to vectors, but you index both rows and columns.

```
grades[
 grades > 85
]
```

	[,1]	[,2]
[1,]	80	90
[2,]	90	90
[3,]	85	85
[4,]	85	100

Matrices

Indexing is similar to vectors, but you index both rows and columns.

```
grades[
  3:4,
  1
]
```

$$
\begin{array}{c|cc}
 & [,1] & [,2] \\
\hline
[1,] & 80 & 90 \\
[2,] & 90 & 90 \\
[3,] & 85 & 85 \\
[4,] & 85 & 100
\end{array}
$$

Matrices

Indexing is similar to vectors, but you index both rows and columns.

```
grades[
  grades[, 1] >= 85,
  2
]
```

	[,1]	[,2]
[1,]	80	90
[2,]	90	90
[3,]	85	85
[4,]	85	100

Demo

Matrices

In this demo, we will:

Index matrices

Knowledge Check

Matrices

Which of the following code snippets outputs the following:

```
[1] 90 100
```

a)

```r
grades[c(1, 4), ]
```

b)

```r
grades[c(1, 4), 2]
```

c)

```r
grades[c(2, 4), 2]
```

	[, 1]	[, 2]
[1,]	80	90
[2,]	90	90
[3,]	85	85
[4,]	85	100

Knowledge Check

Matrices

Which of the following code snippets outputs the following:

```
[1] 90 100
```

a)

```
grades[c(1, 4), ]
```

b)

```
grades[c(1, 4), 2]
```

c)

```
grades[c(2, 4), 2]
```

	[, 1]	[, 2]
[1,]	80	90
[2,]	90	90
[3,]	85	85
[4,]	85	100

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